The Status of Linguistic Constraints in Neural Language Models

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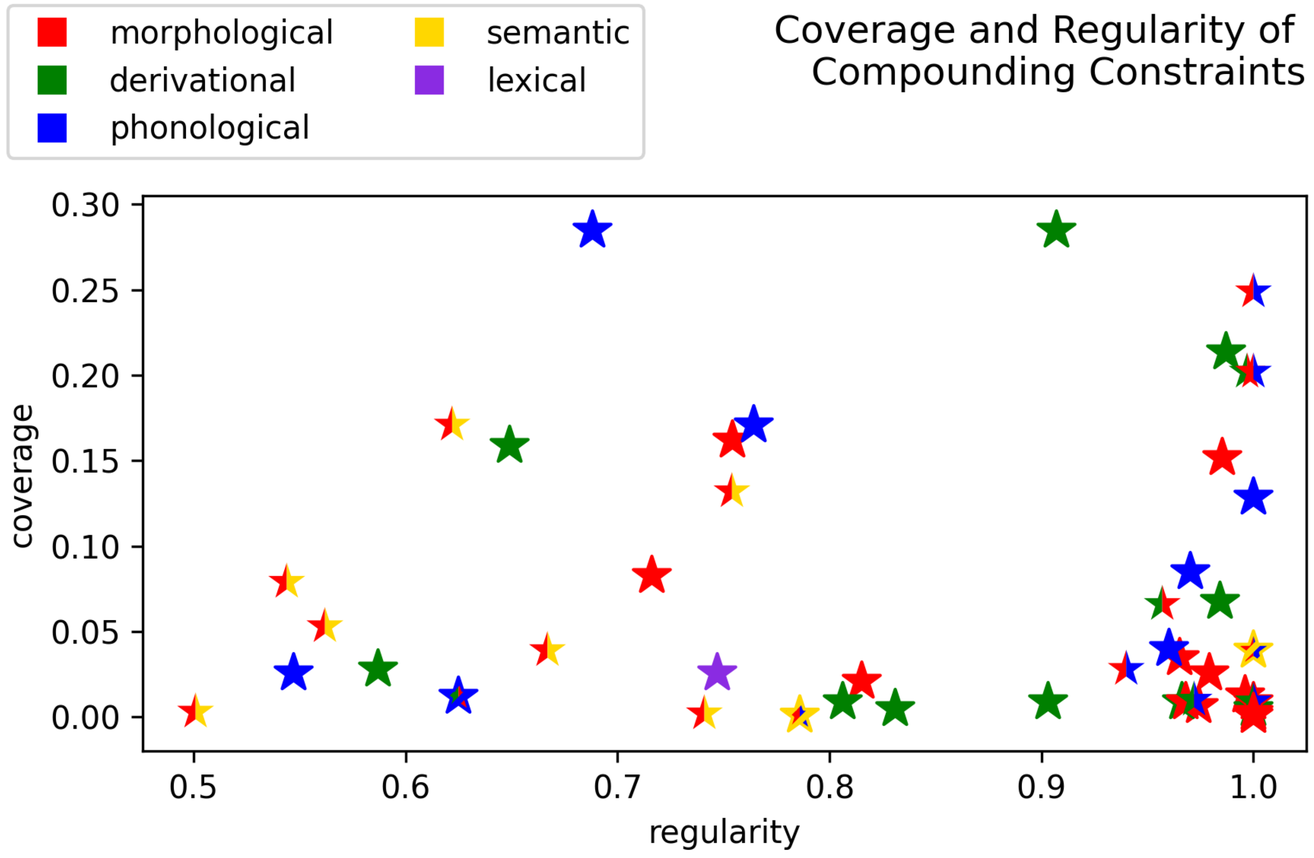
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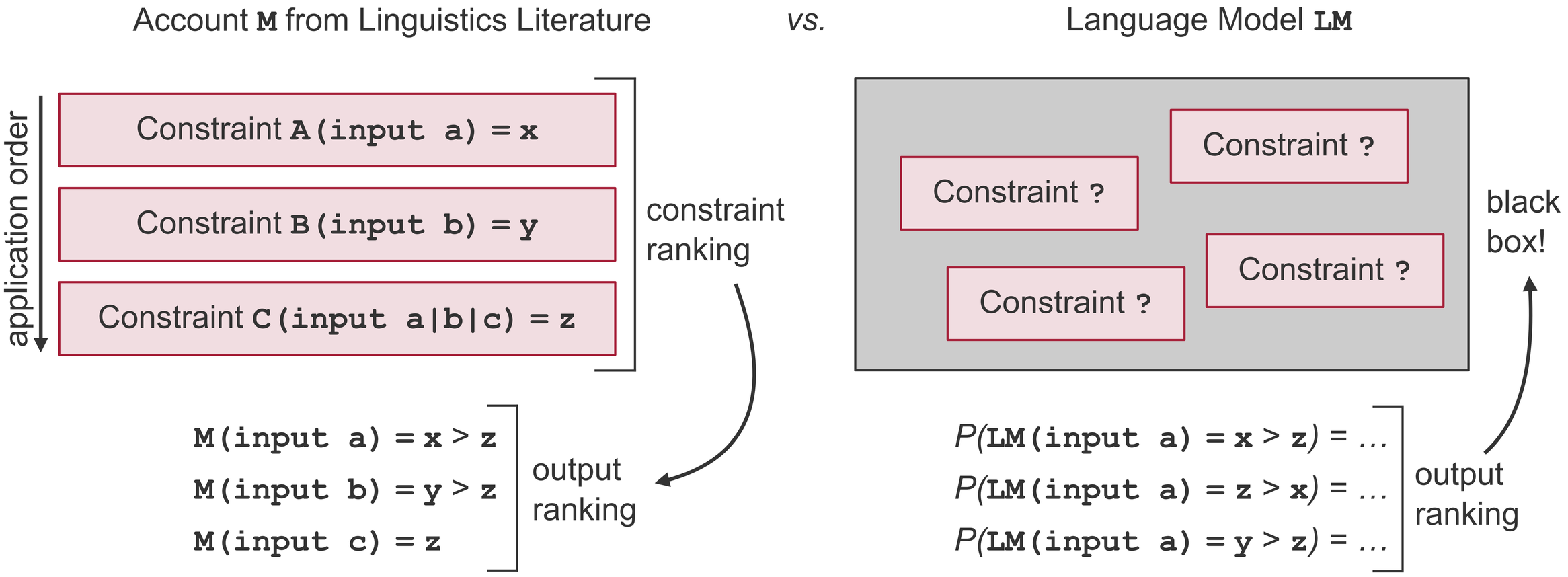
Phenomenon		N Constr.	Description of Constraint Set	Examples of Irregularity
★	Syllabification	< 10	Two central phonological constraints and a number of additional phonological constraints.	hü**gl**ich → hü**.gl**ich *vs.* tä**gl**ich → tä**g.l**ich
★★	Plural formation	10-20	Several central morphological and phonological constraints and a few additional semantic constraints.	Vas**e** → Vas**en** *vs.* Käs**e** → Käs**eø**
★★★	Compounding	> 60	A few dozens of central morphological, phonological, and derivational constraints and a number of additional semantic constraints.	**Punkt** + Spiel → **Punktø**spiel *vs.* **Punkt** + Stand → **Punkte**stand

RQ1: Are LMs able to abstract (weak) linguistic constraints from the training data?

RQ2: If abstraction capabilities are confirmed, how similar are the abstracted constraints to the frameworks established in the linguistic literature?

RQ3: Otherwise, how does explicit insertion of constraints in the prompt change the model generations?





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